

Module 1

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Module 1 Study Guide and Deliverables

Readings:

- Amatayakul, Chapters 1 and 5
- [Updated HIMSS Davies Award Winner Virginia Women's Health Center Application](#) (posted on course Blackboard discussion forum)
- Review - [ONC Workflow Tools](#)
- Review - [AHRQ Toolkit on Workflow Assessment and Re-design](#)
- Optional – [More Detail on Business Process Re-engineering](#)
- Recent article – will be posted in the Live Classroom Slides and Materials discussion forum

Assignments:

Assignment 1 due Tuesday, January 23 at 6:00 AM ET

Assessments:

Quiz 1 due Tuesday, January 23 at 6:00 AM ET

Discussions:

For grading consideration, all posts to this discussion topic must be made prior to Tuesday, January 23 at 6:00 AM ET. Any posts after this date/time will not be included in the grading process.

Live Classroom:

- Wednesday, January 17 at 8:30 PM ET
- Saturday, January 20 at 12:00 PM (noon) ET
- Facilitator Session: Friday, January 19 at 8:30 PM ET

Module Learning Objectives

After successfully completing this module, participants will:

- Understand at a high level what a Health Information System (HIS) and an Electronic Health Record (EHR) are
- Know the major success factors to successful deployment of an HIS/EHR system
- Have a high-level understanding about government incentives and requirements for healthcare organizations adopting EHRs
- Grasp major healthcare clinical workflow processes
- Understand high-level concepts of business process re-engineering (BPR)
- Know how to apply BPR as part of deploying or using an HIS/EHR
- Know what an Enterprise Architecture is and how to use it to understand HIS/EHR functionality
- Know the four layers of the TOGAF Enterprise Architecture
- Understand the benefits of an Enterprise Architecture

Lecture 1: Introduction to Health Information Systems (HIS) and Electronic Health Records (EHR)

How Is Healthcare Delivered in the U.S.?

Healthcare in the United States is delivered through multiple organizations with different but interrelated goals. Think about what happens when you are ill. Often, you start by visiting a primary care provider. Based on the symptoms you show, the provider will order various diagnostic tests or procedures. If appropriate, the provider will refer you to a specialist who may order more diagnostic tests or procedures. These providers are referred to as outpatient providers or ambulatory care providers. If necessary, you may also be referred to a hospital for in-patient or acute care.

In an emergency, you often go directly to an emergency room, perhaps by ambulance, and possibly get admitted into the hospital. In the process of interacting with these various organizations, your providers are interact with many other healthcare organizations, such as laboratories, radiology departments, pathology departments, and others. Alongside these clinical activities, a full series of business activities occurs, including tracking the care you receive and the materials used, and managing billing and insurance claims.

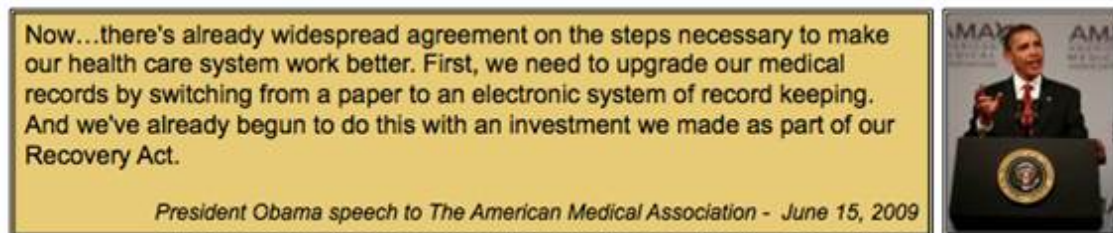
Another characteristic of the U.S. healthcare system is its revenue model. Historically, U.S. healthcare providers have been paid under what is called fee-for-service. Essentially, they are paid a certain amount for the different clinical actions they take. For example, a provider might be paid a certain amount to do one to four stitches and a larger amount for five to ten stitches. Providers therefore are motivated to do “more” to both provide the best care and to get more money. Patients often do not understand the costs—or the benefits—of their healthcare, especially since the full costs are often buffered by their healthcare insurance. The system therefore has limited incentives to moderate costs. The federal government and private insurers are now changing this approach by adopting new models, variously called *managed care*, *risk-sharing*, or *accountable care*, that tie payments to providers to care results and quality.

This complex healthcare delivery process requires substantial paperwork. Healthcare providers and organizations manage a large amount of clinical information—vital signs, symptoms, tests, treatments, and much more. At the same time, they track personal and business information like name and address, insurance coverage, billing information, and other business-related data.

Historically, this information had been stored on paper. Our health information—perhaps the most important data in our lives—was not tracked electronically. Even today, there are times that health information is tracked on paper and, even more frequently, poorly shared amongst the many participants providing care. When we contrast this with the digitization of information in virtually every other area of our lives, from banking to communication to shopping, management of healthcare information has been alarmingly antiquated.

So, What Changed? Federal Government Incentives

Faced with this lack of adoption of EHRs, the federal government decided to act. In the first quarter of 2009, as part of the American Recovery and Reinvestment Act (ARRA), the federal government committed to a major initiative for U.S. deployment of EHRs that both required and helped pay for EHR deployment. This part of the ARRA is referred to as the *Health Information Technology for Economic and Clinical Health (HITECH) Act*. With mandated change and funding coming from this important federal initiative, we entered a new era of adoption of EHRs.



Under the HITECH program, eligible providers and hospitals had to both implement and demonstrate what is known as *Meaningful Use* of a certified EHR system by a certain timeframe to claim incentive payments. Meaningful Use (MU) requirements were essentially the government's quality control measures. They held healthcare providers accountable by requiring that they demonstrate effective use of EHRs. Further, they couldn't use just any electronic method of managing records—they have to use an EHR system that meets federal certification requirements.

The HITECH program is now completed. The federal government has implemented a replacement for the HITECH program and for Meaningful Use as part of the Medicare Access and CHIP Reauthorization Act of 2015 (MACRA). This legislation continues the change of the U.S. healthcare financial model from fee-for-service to managed care. Under this new approach, Medicare payments are regulated either by an Alternative Payment Model (APM) or under a part of the MACRA legislation called the *Merit-Based Incentive Payment System (MIPS)*.

Both incentive providers to use EHRs as a critical part of achieving their goals. The component of MIPS related to EHRs is called *Promoting Interoperability (PI)*.

The HITECH Act and the new APM and PI regulations successfully increased EHR adoption. According to the federal government, as of the end of 2017, over 85% of office-based physicians had adopted a certified EHR and over 95% of hospitals had achieved Meaningful Use. (See [Data on Health IT Dashboard](#)).

At the same time as this increase in EHR adoption, healthcare providers in the United States have found they need more comprehensive information systems that not only computerize clinical information but also computerize other functions such as scheduling, coordination of care across provider entities, analysis of healthcare performance, inventory tracking, revenue cycle management, facilities management, and more. As a result, comprehensive Health Information Systems (HIS) that incorporate EHR and other functionality are increasingly being used. This course covers the deployment and operation of both types of systems referred to as HIS/EHRs.

Balanced Performance Model

Successfully deploying and using an HIS/EHR system—or, for that matter, any enterprise information system—requires balancing several key aspects of the deployment, what we call the *Balanced Performance Model (BPM)*. This balance enables meeting the organization's goals for the HIS/EHR system by managing the deployment successfully. The following diagram outlines these critical areas and how they are integrated.

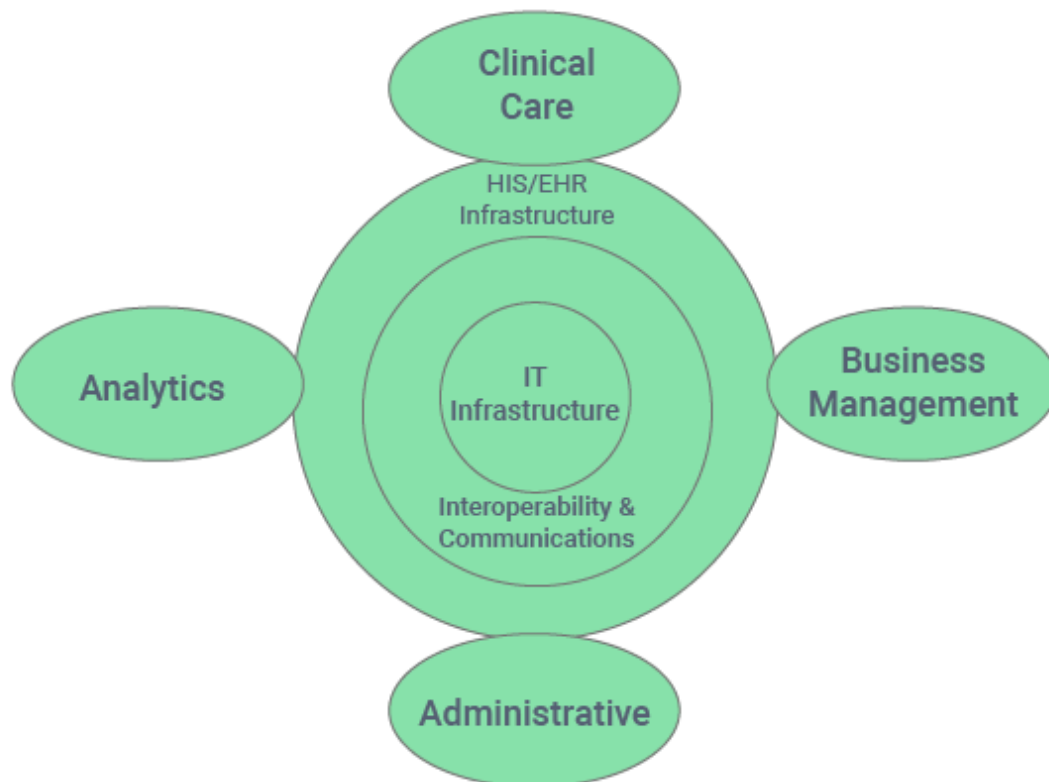


All through this course, we will tie the information in learning objectives to the corresponding items in this Balanced Performance Model. This lecture reviews these areas at a high level and the rest of the course delves into them more deeply.

Technology

We start with the technology component of the Balanced Performance Model (BPM). This component consists of the HIS/EHR system functionality and the related technical infrastructure. The different functional areas of an HIS/EHR system are summarized in this diagram and described below.

HIS/EHR Functionality



HIS/EHR Functionality

To understand HIS/EHR functional capabilities, it helps to know some terminology specific to healthcare. Let's first define different types of care environments.

- Acute: Hospital settings—often called in-patient or acute care providers
- Post acute: Still an in-patient provider but less intense than a hospital, such as a rehabilitation hospital
- Long term: Typically, residential but designed for long-term residence, such as a nursing home
- Ambulatory: Non-residential clinical settings, such as doctors' offices and clinics
- Home: Care provided in the home

HIS/EHR systems are used in all of these healthcare provider environments. Increasingly, the systems are integrated to better track and coordinate care across them. So, HIS have functionality in four broad areas:

- Clinical Care—tracking and managing clinical activities
- Analytics—understanding the health and the effectiveness and cost of treatment of groups of patients
- Business Management—managing the finances and especially the revenue of a healthcare provider
- Administrative Management—managing the non-clinical activities of a healthcare provider

In addition, HIS/EHR systems are increasingly connected, called *interoperability*. Successful interoperability of HIS/EHR systems requires standards and information systems, which will be covered in a later lecture.

Clinical Care Functionality—Electronic Health Records

Clinical care functionality is embodied in the electronic health record (EHR). Its typical clinical functions include housing patient information such as exam information, diagnostic tests, labs, and results. EHRs also have other functions, such as clinical decision support, order entry and management, tracking and reporting features, patient care management tools, alerts, and referral information. There are electronic prescription capabilities, so clinicians can send patient prescriptions to pharmacies securely and instantly online. There are also secure patient portals where patients can do things like access their health information, schedule appointments, and communicate with their doctors.

What Is a Certified EHR System?

As part of the HITECH Act and continuing under PI, the federal government has mandated certain standards that an EHR system must meet to be certified. The goal of certification is to ensure that an EHR has the functionality clinicians need to achieve federal regulations. Use of a certified EHR is required under PI. Note that federal certification is focused on clinical functionality and does not include many other functions of an HIS/EHR.

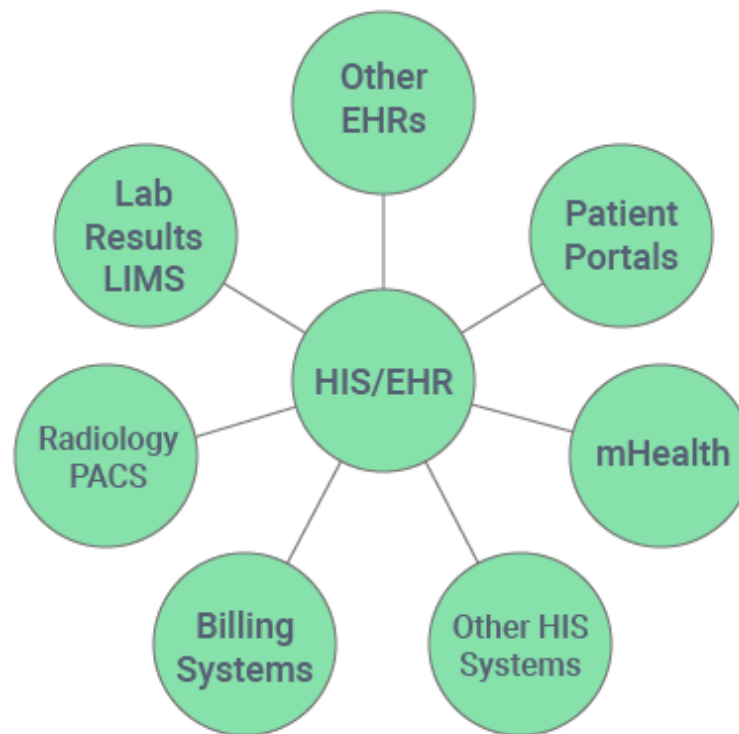
The Office of the National Coordinator for Health Information Technology, abbreviated ONCHIT or ONC, is a department within the U.S. Department of Health and Human Services (HHS). ONC's purpose is to promote a national health information technology infrastructure and oversee its development. ONC has created a program to certify EHRs. We will review this program in depth later in this course. The list of certified EHRs is large and changes frequently.

HIS/EHR Technical Infrastructure

The technical infrastructure is the "behind the scenes" layer that sits underneath the HIS/EHR application layer and is essential to its operation. Liken this to a kitchen—the electricity and gas that power a stove or the hot- and cold-water lines that feed into a sink—are the elements behind the walls that make the kitchen work. In the case of HIS/EHRs, it's the network, servers, security, etc.

The technical infrastructure must provide the capabilities the HIS/EHR needs to run successfully and to enable the connectivity HIS/EHRs have with other systems, as shown in the diagram below:

Technology – The HIS/EHR Environment



When fully adopted, HIS/EHRs are essential for successful delivery of healthcare. Without a working HIS/EHR, clinicians do not have the information about patients' health history or ability to provide care. Therefore, operation of an HIS/EHR requires that this infrastructure operates at a high-performance standard. The requirements to facilitate this kind of infrastructure fall into several categories: distribution, performance, data management, and security and access management.

These requirements apply to many components of the HIS/EHR technical infrastructure, including end-user devices, servers, storage, networks, databases, operating systems, and more.

Like many types of information systems, HIS/EHRs are increasingly being delivered through a Software as a Service (SaaS) model hosted in the cloud. In this approach, the HIS/EHR vendor runs and operates the system in a data center of their choosing that is delivered over the Internet—the cloud—and has responsibility for much of the infrastructure, such as servers and storage. The healthcare organization using the HIS/EHR does not need to provide these parts of the infrastructure, though they do need to continue to provide some technology such as network capabilities to access the HIS/EHR and end-user devices.

HIS/EHR systems are "mission-critical" systems that require nearly 100% uptime. As a result, the HIS/EHR system's operational requirements, including availability, reliability, scalability, portability, usability, and security, are complex. In later lectures, the course will review these requirements in detail.

Security is an especially essential requirement for an HIS/EHR. Again, security is covered in detail in a later lecture, but at a high level, the security principles for the technical infrastructure of an HIS/EHR involve five

layers:

- Integrity
- Availability
- Alteration
- Disclosure
- Destruction

People

The second element of the BPM is the people element.

There is a diverse mix of people involved in the use and deployment of an HIS/EHR system, including clinicians, administrators, technical, and other staff, and the patients themselves. Every person involved in the world of healthcare uses an HIS/EHR and depends on the data it generates. So, an effective deployment program must gain the support and adoption of this broad base of users. The program must include education and communication that provides these individuals the knowledge they need to successfully use the HIS/EHR.

In addition, support is an essential part of the successful deployment of an HIS/EHR. These are highly complex, mission-critical systems involving many people and a tremendous amount of data. Problems are inevitable. To ensure the business of healthcare continues, there must be quality and responsive support to address problems expediently, keep the system operating and available, and help users with the system. The support must function during implementation, and after, it must be effectively managed and well organized.

Process

The third element of the Balanced Performance Model is the process element. The rethinking of processes to function in the new environment of an HIS/EHR must be methodical and well planned. A methodology known as *Business Process Reengineering (BPR)* is used in this element. Our course will cover BPR in the next lecture.

Business

The fourth element of the Balanced Performance Model is the business element. Healthcare is a business and HIS/EHRs must meet multiple business requirements and constraints. When a healthcare organization deploys and operates an HIS/EHR, it must understand the costs and potential increased revenue from using the system, define an affordable budget, and manage deploying and operating the HIS/EHR within that budget. Success also means understanding other, non-monetary business aspects. In addition, the successful deployment of a HIS/EHR must address several important business requirements, such as managing a complex project and selecting an HIS/EHR system and vendor and then working with that vendor.

What Is the Value of HIS/EHRs?

The BPM provides a framework for successfully deploying a HIS/EHR system. But we also need to understand the benefit of HIS/EHRs. They impact the quality of care as well as costs and revenue.

The federal government recognizes the value of EHRs and reflects that value in the Meaningful Use and PI regulations. The regulations partly tie federal payments to EHR use through these regulations. These programs will be discussed later in this course.

Federal EHR Requirements Focus on Five Major Areas

1. Improve quality, safety, and efficiency, and reduce healthcare disparities
2. Engage patients and families in their healthcare
3. Improve care coordination
4. Ensure adequate privacy and security protection for personal health information
5. Improve population and public health

The federal requirements differ depending on the type of healthcare organization, whether a hospital or an ambulatory provider. Some requirements include:

- Eligible providers need to prove that they have met different objectives with their use of an EHR to be considered “meaningful users” under the PI regulations. These objectives change regularly but can include requirements like computerized physician order entry (CPOE), the use of clinical decision alerts, incorporation of lab results into the EHR as discrete data, ePrescribing, and electronic information distribution to patients.
- Clinical quality measures need to be achieved and submitted by providers on a core set of measures as well as specialty specific subgroups. These requirements change regularly.
- Under the PI component of MIPS and similar requirements of APMs, adjustments to Medicare reimbursement are made based on achieving specific requirements.

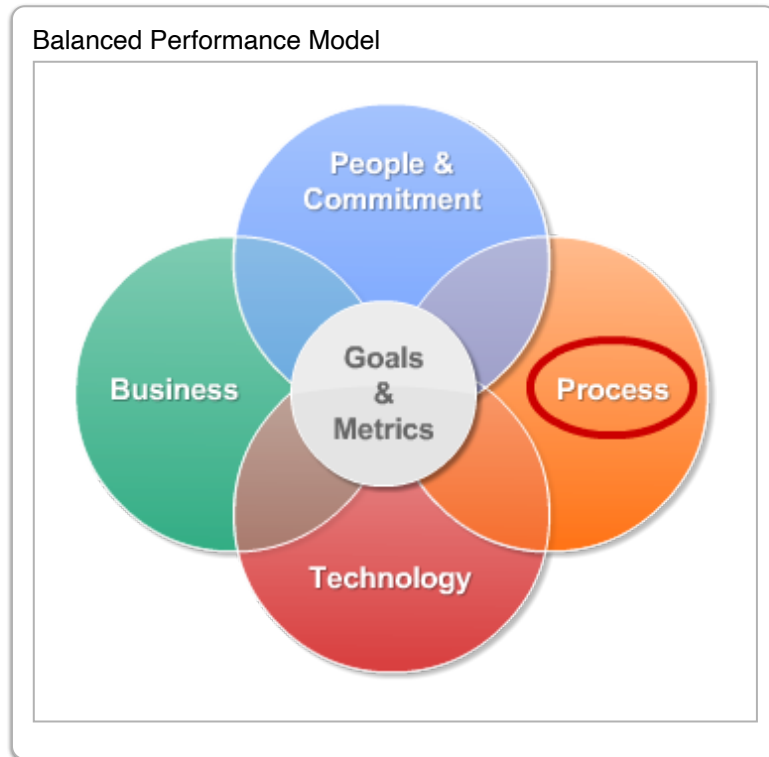
Summary

The success of an HIS/EHR deployment requires attention to each element in the Balanced Performance Model. Ultimately, success is measured by clinical quality and financial results. We will use the BPM as a guide throughout this course to help structure an HIS/EHR deployment program.

Lecture 2: Healthcare Workflow and Business Process Re-Engineering

Overview

As previously discussed, implementation of an HIS/EHR system is a complex process that requires successful integration of all components of the Balanced Performance Model.



One factor that makes HIS/EHR deployment complex is the number of people involved in healthcare delivery. Healthcare is ultimately about people helping people. The California Healthcare Foundation states in a study: “Health care workflows tend to be quite complicated because they are carried out by a variety of staff members, departments, and care partners. Documenting these processes can lead to a detailed understanding of how different tasks interact and depend on each other. Often multiple processes need to be coordinated, with time-sensitive information handed off to different departments and provided to patients.” (Source: California Healthcare Foundation “Workflow Analysis: EHR Deployment Techniques,” January 2011)

More broadly, whenever a complex activity is automated, the focus must be on workflow. The dictionary defines workflow as follows:

work•flow

Noun: The sequence of industrial, administrative, or other processes through which a piece of work passes from initiation to completion.

Using HIS/EHRs impacts healthcare workflow in a variety of ways. The first step in implementing an HIS/EHR system is therefore to understand and adapt the process element of the Balanced Performance Model. In so

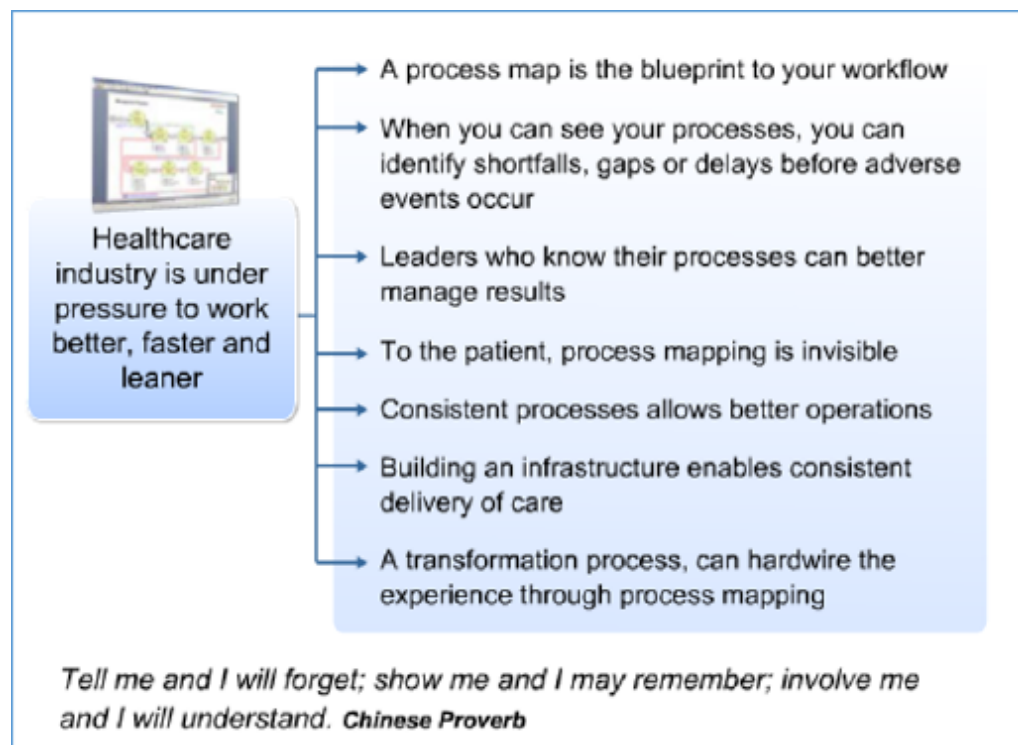
doing, we will cover:

- What healthcare staff (clinicians, administrators, etc.) do on a daily basis,
- How an HIS/EHR will affect these processes,
- What an HIS/EHR does, and
- How it can improve workflows.

The goal of an HIS/EHR is to optimize the strengths of people, processes, and technology together rather than to simply digitize paper tasks or automate them. A paper-based workflow will not serve as a successful blueprint for a digital workflow. It takes careful process analysis and planning to transition from paper to HIS/EHRs and to successfully use an HIS/EHR. The same is true when upgrading to a new HIS/EHR. It's not as simple as just deploying technology. It's re-engineering and optimizing processes and configuring and even customizing the technology to enable clinicians to deliver excellent care.

Teams must commit to auditing the tasks they perform in every aspect of their jobs to create a detailed record of the As Is process. They must also formulate a goal for how a process can be improved with the use of an HIS/EHR system. Finally, the two must be compared and the gap between them clearly defined. Only then can an effective re-engineering take place.

Importance of Improving Process Workflow



A process map is the blueprint to any workflow. It allows all participants, including clinicians, administrators, and patient representatives, to clearly see the processes, which helps them more easily identify problem areas or the need for changes. It allows the participants in the process to understand where events are first triggered and how to better manage results. It is a technique to optimize efficiency, productivity, and safety. The patient receives the care or service as defined by the workflow. So, appropriately involving patients in process mapping is extremely valuable.

Processes are often inconsistent among different sites and even different staff within one site. Standardizing these processes and making them consistent can significantly improve every aspect of an organization's operations. Incorporating these standardized processes into a technological infrastructure enables much better access to unified information and consistent delivery of care.

The outcome of process mapping of an existing workflow can sometimes show that mistakes are made due to lost information and missed documentation. In healthcare, this can result in a care provider being a "reactive care" organization. When processes are re-engineered as part of implementing an HIS/EHR system, these same organizations can be transformed so they provide better quality of care.

This is exemplified by HIMSS Ambulatory Davies Award Winner, James F. Holsinger, MD, PC.

“We felt strongly that an EHR system would assure a higher quality of patient care than a paper system. There would be no lost records, no missed documentation and we could program in quality measures such as drug interactions and preventive care reminders. If we chose the right system, we could achieve our long-term vision of preventive care versus reactive care.”

Business Process Re-engineering

The work to redo and improve healthcare workflow or processes exemplifies Business Process Re-engineering (BPR). The remainder of this lecture introduces BPR.

For every workflow in an organization, there are three different processes:

- The perceived process—what the work force or organization thinks the process is
- The reality process—what the process actually is or the As-Is process
- The ideal process—what the process Should Be to generate the best outcome

When mapping a process, it's important to start with the process that is really being followed or the reality or As-Is process. This is the only way to correctly identify gaps and to plan the ideal process with automation accurately. One possible business process re-engineering error is to start with the perceived process. Since it does not reflect what is really happening in the organization, the newly defined process can be sub-optimal or even wrong.

Every process, whether within or outside of a professional organization, has these three versions. An example of this is doing the laundry.

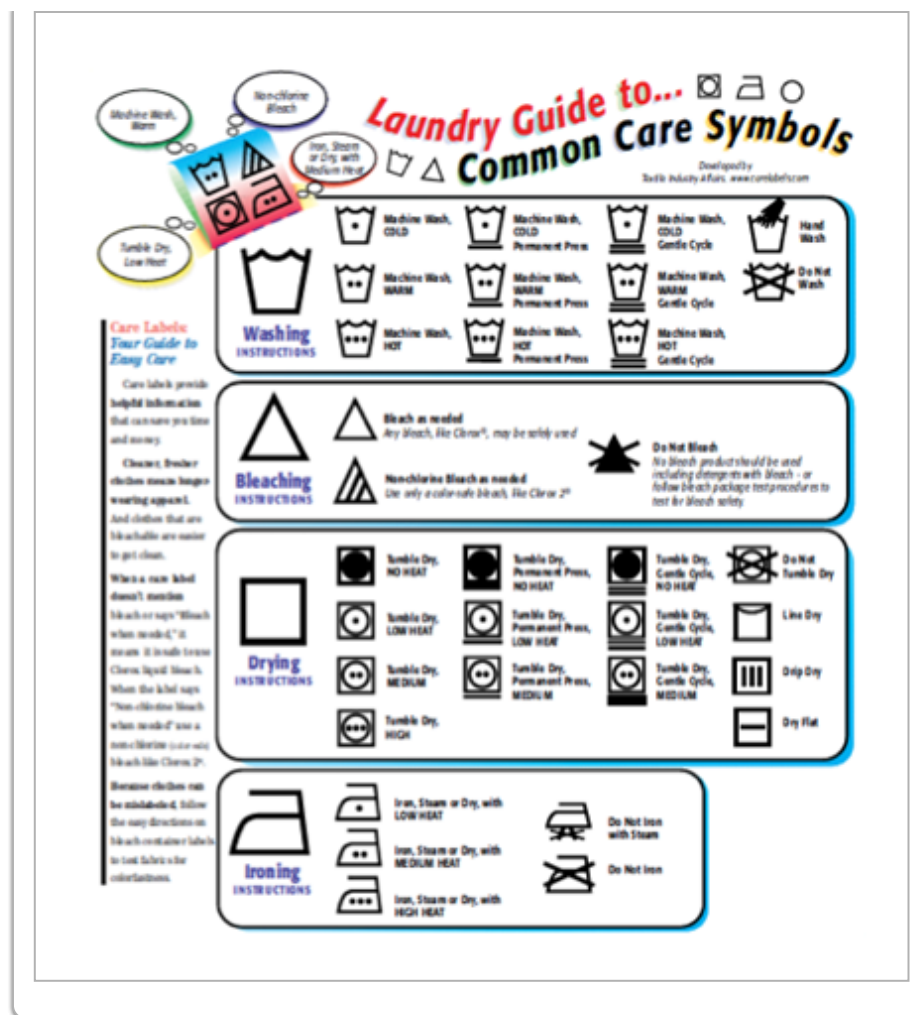
The Laundry Process



When doing the laundry, you might put the laundry in the hamper, sort it by color and fabric type, wash it according to its type, dry it, and fold it. But, every person has a different way of doing the laundry. One person might not sort it into colors but rather into lights and darks. Another person might just put it all together and press start, not checking the settings on the machine.

The garment industry provides us with an ideal process—the perceived process as shown in the laundry guide below.

The Laundry Process



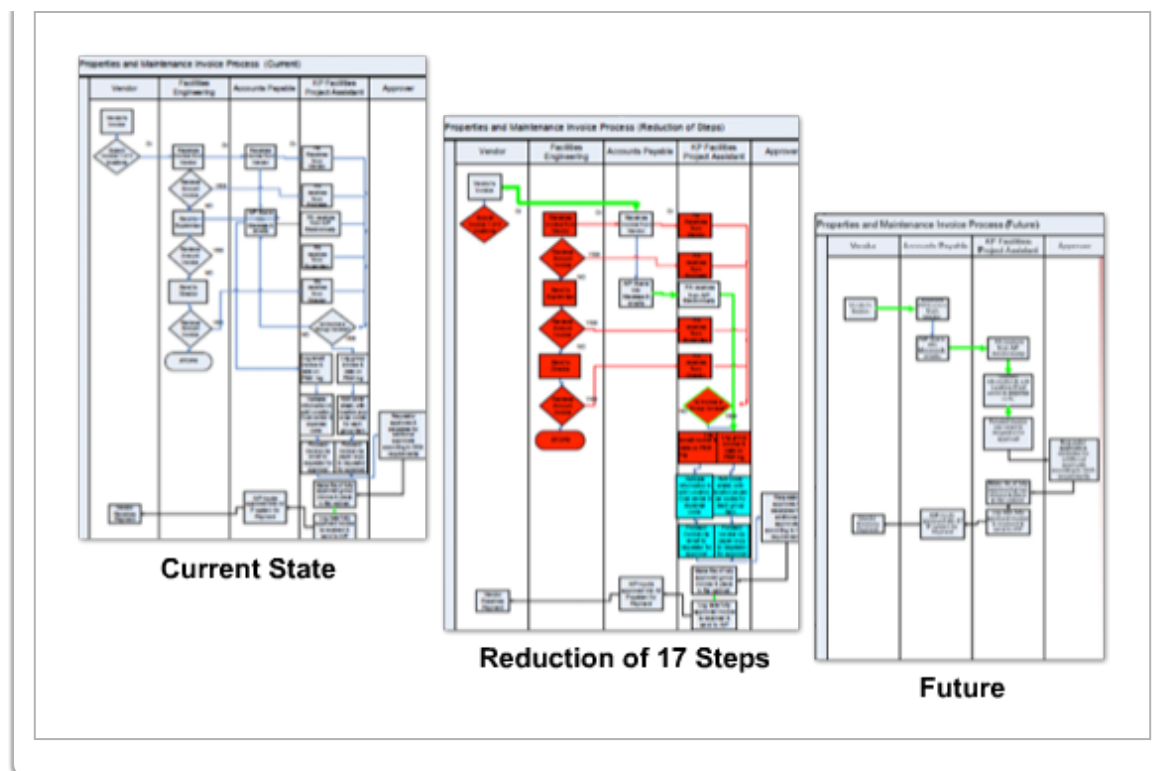
Even though most people might think that they follow these care instructions (the perceived process), they actually do it quite differently (the reality or As-Is process) and may not even know what these symbols mean. So, each version of the process for doing laundry is:

- What we're told we should do by the care instructions,
- What we really do, and
- What we ideally should do under an improved process.

This same kind of thinking can be applied to a healthcare organization doing BPR when implementing an HIS/EHR.

There are several basic steps involved in the creation of a process map.

Importance of Improving Process Workflow



First, you map the current state or the s-s process. Then that current state is reviewed with a goal to make it better. Typically, that review will result in a reduction of steps in the process. Finally, a new future state or ideal process is defined that better achieves the goals of the process and can often make it more efficient.

Benefits of Doing BPR as Part of Implementing an HIS/EHR

BPR can be a complex, time-consuming exercise. So, why do it? The benefits of BPR coupled with the benefits of implementing an HIS/EHR system can be significant. The new processes or workflow should reflect the use of the HIS/EHR and be an improvement. An improvement could be better care quality, increased health of patients, cost reduction, or a mix of all of these.

Fundamentally, the best optimized process is “better, faster, and cheaper:”

- Better results—better quality patient care and easier for the participants
- Faster—patients get care more quickly or more patients can be treated
- Cheaper—costs are reduced

But doing BPR might not result in all three of these. For example, better care quality might be at a higher cost but deemed well worth that cost.

A HIS/EHR is a tool that automates healthcare processes—clinical and other—and helps improve them. The benefits of BPR and related use of a HIS/EHR include:

- Increased effectiveness of the work—better quality care delivery
- Increased efficiency—reducing time required, providing quicker response
- Reduced cost of the work
- Reduced risk—safer healthcare—an example is including drug interaction alerting functionality in an EHR
- Increased flexibility and adaptability to changing needs
- Better jobs—improved staff morale, less turnover
- Improvement in overall organizational results delivering healthcare

These benefits are illustrated in the following video, in which Dr. Eric Hartz, Chief Medical Information Officer (CMIO) at Eastern Maine Medical Center and the medical director for their meaningful use project, discusses the benefits of using an EHR system, provides tips for participating in the EHR Incentive Programs, and explains how they implemented EHRs across several hospitals.

EHR: Dr. Eric Hartz's EHR Story from the 2012 HIMSS Conference



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EHR: Dr. Eric Hartz's EHR Story from the 2012 HIMSS Conf...



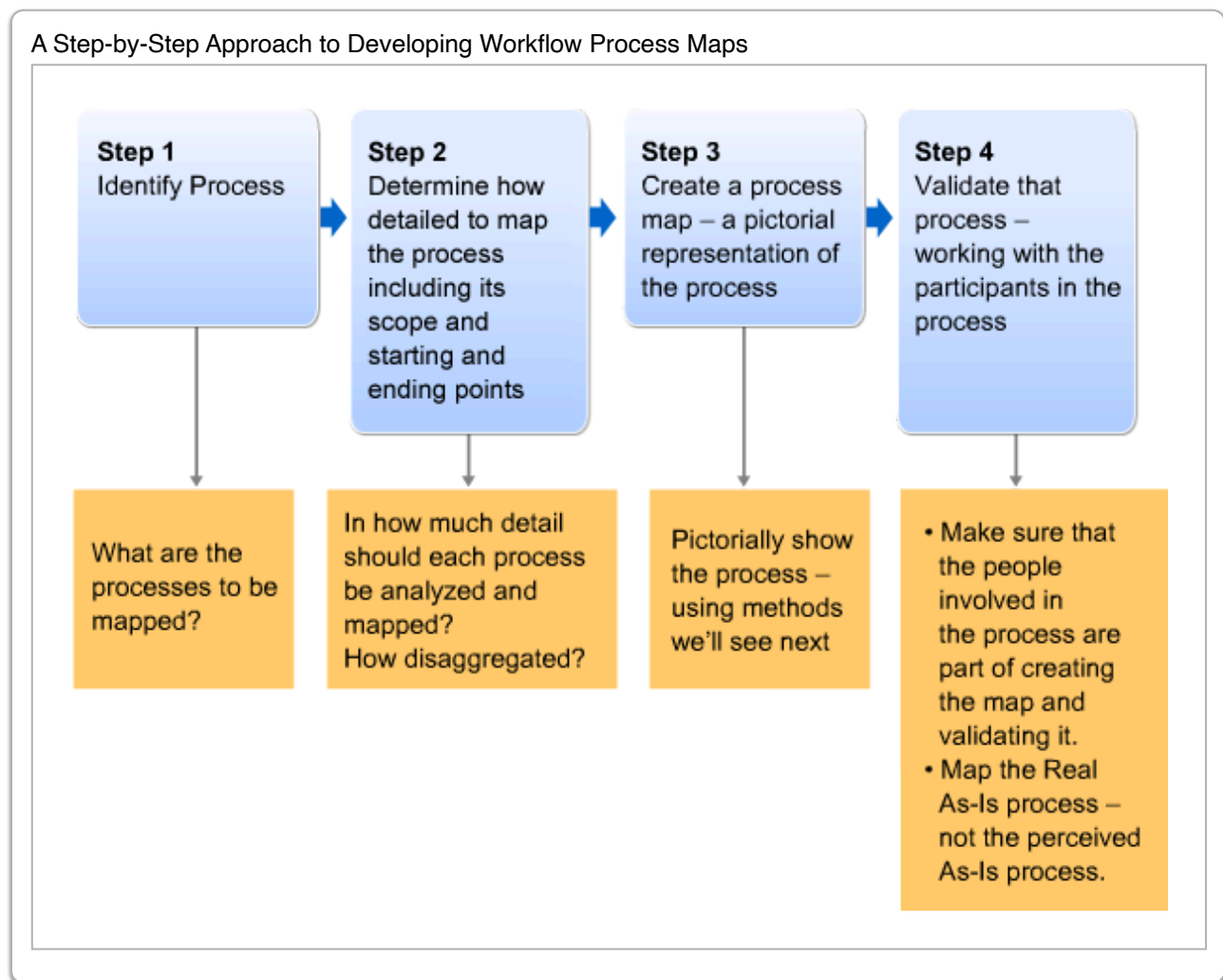
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Dr. Hartz had always been aware that implementing an EHR system and standardizing processes would be a necessary step toward achieving strategic goals, including patient safety and financial concerns. With the goal of having every eligible provider in the state using an EHR, a network-wide project was created, with teams of people committed to finding the resources necessary to implement EHRs and standardize processes across departments and sites.

Dr. Hartz talks about how merely digitizing processes doesn't necessarily mean there is a benefit to patient safety. Processes must be redesigned. Also, patients feel more confident in their care when they see a concerted effort among their providers using a centralized digital system instead of a paper file.

How to Do BPR

Successfully performing business process re-engineering involves a step-by-step approach to understanding workflow by developing workflow process maps.



Process mapping is often done using flowcharts. Start by representing the first task in the workflow with a symbol or box and writing a description of the task in the box. Beneath the symbol, note the names of people or position titles involved in doing the task as well as the departments involved.

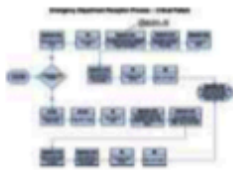
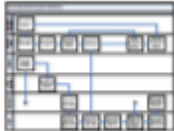
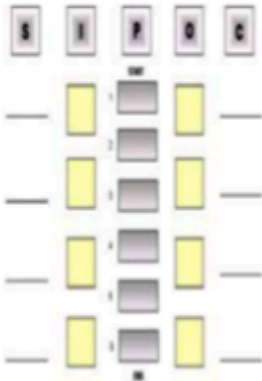
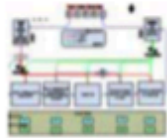
Draw a second symbol for the next task in the process and connect the two with an arrow indicating the direction of flow.

Continue this way until all the tasks in the process are documented. Validate the process by walking through the tasks and completing the workflow as documented.

Each symbol should represent only one action or decision. Determining whether the task is complete should be posed as a yes/no question. For example, "Is the laundry sorted?" If yes, move onto the next step. If no, repeat the prior task.

BPR is a well-developed field, which, when necessary, can be very complicated and detailed. In these more sophisticated approaches, there are several types of workflow diagrams representing process maps that range from high level to very detailed. The following are some examples. More detail is in Chapter 5 of the textbook, *Health IT and EHRs*.

Developing Workflow Process Maps: Type of Maps

Maps	Purpose	Example
High-Level Process Map	A high-level process map displays only the most important steps in a process. This type of map usually contains six to eight process steps.	
Detailed Process Map	A detailed process map depicts much more information regarding all of the activities that occur within the process. This variety of map may show where decisions are made, waiting periods, tasks that frequently require rework and feedback loops.	
Deployment Process Map	A deployment process map, also known as a swim lane diagram, provides information about the specific departments or individuals who are responsible for performing specific activities within the process.	
SIPOC/COPIS	A high-level process map that graphically displays the major events occurring in the process. SIPOC stands for: Supplier: a person or organization who provides the inputs to your process. Input: materials, resources and data required to execute your process. Process: a series of steps or activities that uses one or more kinds of inputs and changes them into an output that has value to the customer. Output: a tangible service that results from the process. It is equivalent to what a customer deems critical to quality (CTQ). Customer: a person or organization who receives the outputs of the process. It can be either an internal or external customer.	
Value Stream Map	A value stream is used in the LEAN Process Improvement methodology and helps you understand the flow of both material and information. Tends to have five to ten process steps.	

There are also several common workflow improvement methodologies that organizations can implement when approaching BPR. Some include:

- Total Quality Management (TQM)
- Plan-Do-Check-Act (PDCA)
- Six Sigma

- LEAN Systems
- LEAN Six Sigma

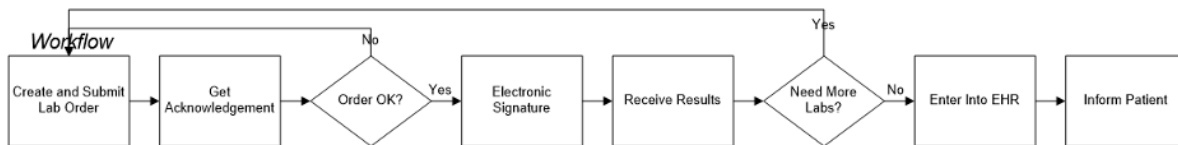
Many of these methodologies originated in the defense industry or in engineering and manufacturing organizations that have very complex and sophisticated processes to document and improve upon. While healthcare also has complex processes, the detailed, in-depth approach of many of these BPR methodologies can be more than is necessary when implementing an HIS/EHR. This is especially true of doing BPR for small, ambulatory care providers. In these cases, a simple, straightforward flowchart developed, as shown below, will be sufficient.

Therefore, in this course, we will do business process re-engineering and workflow mapping using a simpler five-step methodology. The five steps relate to answering five core questions when mapping processes for any given activity:

- What is the workflow?
- What is the key information needed at each key process step?
- What people in the practice are involved for each process step?
- What are key HIS/EHR functions that can be used for each process step?
- What are possible benefits from changing/improving the workflow?

Diagrammatically, this BPR process and mapping can be shown as follows. This example shows a high-level process for ordering a diagnostic test.

Diagnostic Test Ordering - Example BPR



Data Needed

•Diagnosis •Possible labs	•Lab acknowledgement	•Original Lab Order •Lab Ack. •Diagnosis	•Physician ID	•Original Lab Order •Lab Results •Diagnosis	•Original Lab Order •Lab Results •Diagnosis	•Lab Results •Diagnosis	•Patient Info •Lab Results •Diagnosis
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Who is involved?

•Physician •Nurse •Lab	•Physician •Nurse •Lab	•Physician •Nurse	•Physician	•Physician •Nurse •Lab	•Physician	•Physician •Nurse •Admin Staff	•Physician •Nurse •Patient
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EHR Functionality

•Order Sets •Order Protocols •Lab System (LIMS) Interface	•Lab System (LIMS) Interface	•Clinical Decision Support •Order Protocols	•Electronic Signature	•Lab System (LIMS) Interface	•Clinical Decision Support •Order Protocols	•Lab System (LIMS) Interface	•Patient Communication
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Benefit

•Faster •Less Error Prone	•Faster •More available	•Faster •Less Error Prone	•Safer •Faster	•Faster •Greater availability	•Faster •Less Error Prone	•Faster •Greater availability	•Faster •Greater availability •Improved Communication
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We will use this BPR approach throughout the course. To help you learn and understand it, we include a course assignment on BPR/workflow.

In summary, process maps and workflow diagrams can be a powerful tool for examining a process. Doing BPR helps the organization understand where improvements in processes can be made, the impact of a new HIS/EHR, and how to achieve optimal results. The picture you create when building a detailed process map will quickly reveal which steps are useful in getting to the end goal. It is an essential step toward improving the process and identifying the future state or ideal process and to successfully deploying a new HIS/EHR system.

HIS/EHRs and Healthcare Workflow

It's evident from the readings and examples so far that clinicians are realizing benefits by implementing HIS/EHRs and undergoing process mapping and process re-engineering as part of that effort. But what specific aspects of healthcare activities are most affected?

The Agency for Healthcare Research and Quality identified six major categories of EHR impact on workflow:

- Reminders, alerts, and reports that, when provided, increased efficiency and improved workflow

- Administrative and clinical workflow and work/job design that have positive and negative impacts on workflow and job content
- Patient-provider consequences of the health IT system that affected workflow
- Interface design that affected job content and workflow
- System integration that, in all cases, positively affected workflow
- Planning activities and their impact on workflow

As you can see, the processes touch every aspect of what it takes to deliver effective patient care, from administrative tasks to patient interactions. Likewise, every part of delivering healthcare must be reviewed and have its processes mapped as part of this effort so that the delivery of care is well coordinated. Therefore, every department within a healthcare organization and other organizations will be involved. For example, a process map for a hip replacement surgery will involve the pharmacy department for filling prescriptions and the physical therapy department or practice for post-surgical care, but neither the pharmacy nor the physical therapy group will be involved in the entire process.

Healthcare Clinical Processes

As BPR gets more detailed, the minutia can be overwhelming, so let's pull back and simplify things a bit.

Let's start by focusing on ambulatory care since most of us have more experience with that aspect of care delivery. But the BPR approach we use in this course can be used with any aspect of care, such as acute care delivery. In ambulatory care, we can separate processes into two broad categories:

- **Clinical:** The clinical processes of delivering care to patients and managing patient records
- **Business:** The practice management and practice operations functionality

As described in the book *Electronic Health Records*, edited by Jerome Carter, typical ambulatory clinical processes include:

- Registration/Check-in
- Schedule
- Clinical encounter
 - Patient examination
 - Laboratory orders
 - Laboratory results review
 - Electronic prescribing
 - Referral
 - Patient encounter documentation
 - Disease management
- Patient education
- Billing

- Patient check-out

It's important to keep in mind that the typical doctor's visit in an ambulatory setting is an integration of these categories. So, when a patient visits the doctor, a series of tasks occur. To map this process from start to finish, it is helpful to document each task and assign it a category.

For example:

Doctor Office Visit Process Steps	
Area	Task
Check in	Make appointment
Check in	Look up patient information
Check in	Update patient information
Check in	Verify insurance
Check in	Co-pay
Intake	You wait
Intake	Nurse comes and gets you
Intake	Takes you to exam room
Intake	Takes vitals
Intake	Enters into health record—electronic or paper chart
Intake	Gets symptoms
Intake	Exam process
Intake	Nurse leaves
Intake	Wait again
Exam	Doctor looks at record
Exam	Review symptoms, vitals, etc. (SOAP)

Exam	Do tests and/or do orders (lab, radiology, etc.)
Exam	Get results via fax, electronically, by phone, or other
Exam	Review results
Exam	Diagnostics
Exam	Treatment plan
Checkout	Inform patient
Checkout	Follow-up actions (schedule follow-up visit, referral, prescription, etc.)
Checkout	Billing

As described earlier, there are many formal business process re-engineering methodologies. In this course, we will do business process re-engineering using the five-step methodology shown earlier.

The course assignment asks you to use this five-step methodology to do a workflow for the process of referring a patient to a specialist. The assignment gives you hands-on experience doing BPR.

BPR Critical Success Factors

Business Process Re-engineering is a critical and important part of implementing an HIS/EHR or optimizing its use. But it can be complex and hard to do right. Attention to a few critical success factors can help assure it is done well.

1. Recognize that BPR can be hard—it's about change. Change can be time consuming and painful for people. It requires openness and commitment.
2. Be prepared to invest in BPR—it has a cost. It will be mostly a time investment, but it can also cost money for materials, consultants, etc.
3. Top management support is critical. Top managers must be personally involved and lead the project. There will be resistance from some employees. Without the clear and proactive support of top management, success won't be possible.
4. It's all about the people.
 - Listen to their concerns
 - Engage staff in the process—use a diverse and complete team
 - Get the right leader for the team—someone who knows BPR, knows healthcare, is trusted, and can facilitate change successfully

- Be prepared for turnover and attrition—if necessary, allow less versatile and adaptable staff members to leave so you can build the best team for change and realize the full benefit from BPR and the HIS/EHR implementation
5. Use the right methodology—there are many BPR methodologies. Pick one that works. Keep it simple.
 6. Integrate BPR as part of deploying or managing your HIS/EHR. BPR is an essential component of successful use of an HIS/EHR, so do BPR thoroughly and carefully to get the most benefit from an HIS/EHR system.

■ Lecture 3: Enterprise Architectures and HIS/EHR Clinical Functionality

Introduction to Enterprise Architectures

“We have an Enterprise Architecture—
We just can’t show it, explain it, share it,
tell you what’s in it, what it supports,
or who has what.” —Herding Katz

There is a widely used set of methodologies that is valuable to understand the capabilities of an information system and its underlying technical infrastructure. This approach is called an "Enterprise Architecture."

An Enterprise Architecture (EA) ties a business vision, strategy, and activities and aligns them with supporting information systems. The scope of the EA includes the people, processes, information, and technology of the enterprise, and their relationships to one another and to the external environment. An EA is a way of organizing an enterprise’s business processes, information needs, and supporting technologies. An EA is an application of our Balanced Performance Model as it is about using information systems to support the activities of an organization.

EAs are often closely related to and often incorporate business process re-engineering (BPR). They often focus on the processes of an organization, how those processes should change to meet new needs, and the related change management to implement those changes. Then they define the technology to support these new processes. At the same time, new technology can also drive business change as the technology enables new ways to do business. For example, the advent of high-capacity wireless technologies and small, powerful computers—like cell phones—has enabled new, mobile ways of shopping. EA design therefore often takes into consideration four elements:

- The current state (As-Is) process model
- The vision of the future state (To-Be) process model

- The transition plan of the business
- The technology

There are many examples of this type of approach in other areas:

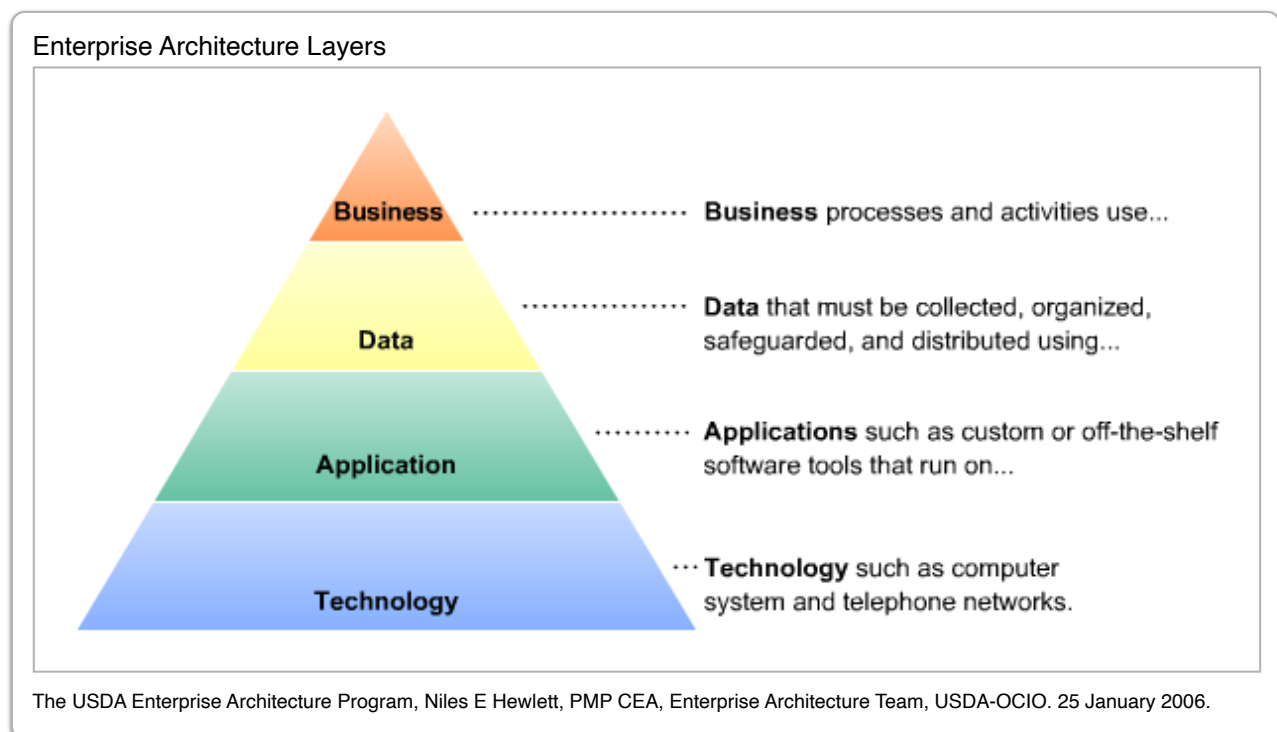
- A blueprint is required before building a house
- Aircraft systems design plans are required to build an airplane
- A community infrastructure plan is required to implement civil engineering for a town—electricity, water, gas, roads, telephone, etc.

In all these cases, the “architecture” links together all elements needed to successfully achieve the goal—whether it is to build a house or an airplane or provide community services.

An EA is not merely an inventory of the technology. It is a methodology for managing the complexity of the technical infrastructure and aligning it with business and process goals. An EA integrates the elements of the infrastructure and aligns them with the business needs—in the case of an HIS/EHR, with delivering healthcare.

Implementing a HIS/EHR requires this type of integrated analysis and, as we have seen, requires revising the work processes. At the same time, changes in healthcare delivery and business are also forces for process change and new uses of technology. The two forces combine when implementing a HIS/EHR, and therefore use of an EA methodology is highly beneficial to successful implementation and use of a HIS/EHR. For these reasons, the BPR process described in this course uses EA concepts.

There are multiple EA methodologies. In this course, we will use one of them: The Open Group Architecture Framework, best known by its acronym, TOGAF. The TOGAF framework divides an EA into four related categories:



The four elements that make up the EA can be presented in a pyramid, demonstrating the hierarchy of each. They include, from top to bottom:

- **Business**—The processes and activities required to operate the business. This is part of BPR and the movement from as-is to should-be processes. For example, with an EHR, an example is the process a patient goes through when visiting a physician.
- **Data**—All of the information that the business processes require, and which must be stored, organized, protected, and distributed securely. In the case of a patient visiting a physician, it is the information about the patient—name, symptoms, diagnosis, treatment, etc.
- **Application**—The application system(s) that use or create the data. In our case, it is the HIS/EHR application.
- **Technology**—The underlying computer infrastructure and networks required to run the application. Like any application system, an HIS/EHR requires servers, networks, etc. to operate.

More information on the TOGAF EA methodology and its related development methodology, ADM, is in this video from Boston University's Course MET CS783 Enterprise Architecture.

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Benefits of an Enterprise Architecture

An EA is beneficial to present and support the current and future vision of a business and the related as-is to should-be process re-engineering and how technology is used to support the business processes. An EA also

helps to support quality decision-making, such as investment choices, and to manage the impact of changes on the organization.

EAs align closely with an organization's mission and goals and the goals of key stakeholders, both internal and external to the enterprise. The EA works to combine the technology, systems, business, and market options to fulfill the enterprise mission, taking into consideration the:

- External environment—Regulations like MACRA/MIPS
- Mission of the healthcare organization—A large, metropolitan teaching hospital has different needs from a small private practice in the suburbs.
- Business strategy (such as emphasis on particular patient populations or diseases)
- Business models (e.g., transformation to shared financial risk business models like accountable care organizations)
- Technology (including existing and new technologies like a cloud-based HIS/EHR)

If an EA is successfully defined, implemented, and followed, a broad and impactful set of benefits should be achieved.

The benefits of an EA include:

- A more efficient IT operation:
 - Lower software development, support, and maintenance costs
 - Reduced redundancy and reuse of existing systems
 - Increased portability of applications
 - Improved interoperability and easier system and network management
 - Improved ability to address critical enterprise-wide issues like security
 - Easier upgrade and exchange of system components
- Better return on existing investment and reduced need for future investment:
 - Reduced complexity in the IT infrastructure
 - Maximum return on investment in the existing IT infrastructure
 - The flexibility to make, buy, or out-source IT solutions
 - Reduced overall new investment lower total cost of IT ownership
 - Leverage new technology
- Better use of new technology:
- Faster, simpler, and cheaper procurement:
 - Buying decisions are simpler because the information governing procurement is readily available in a coherent plan
 - The procurement process is faster—maximizing procurement speed and flexibility without sacrificing architectural coherence
 - The ability to procure heterogeneous, multi-vendor, open systems

